

Tutorial II – Drag

During atmospheric flight, energy is continuously dissipated to the ambient air by aerodynamic drag. Estimation of drag characteristics of a novel aircraft can support the design process, by providing environmental, performance and economic figures at an early stage. The main drag properties can be derived by equations, based on the configuration layout and the expected operating conditions.



Basics

-  What types of drag exist?
-  Give the equation for the total drag force D and also the overall drag coefficient C_D .
-  Name three approaches for estimating aerodynamic drag. What challenges and limitations apply to each method?

Calculation tasks

- 1) Calculate the airspeed V at $M = 0.8$ and determine the dynamic viscosity μ as well as the density ρ of air at an altitude of $h = 10\text{km}$.
- 2) Determine the equivalent parasite drag areas f_j of the individual assemblies (fuselage, wings, empennage and engines) at the flight conditions of task 1). The assemblies are constructed of polished sheet. Neglect interference drag and assume fully turbulent flow over all assemblies.
- 3) Now derive the total parasite drag area f from the individual parasite drag areas previously calculated. Finally, derive the zero-lift drag coefficient C_{D0} of the Airbus A320 from this value.
- 4) Determine the relation between the coefficient of induced drag C_{Di} and the lift coefficient C_L for this aircraft in clean configuration. Also, draw the drag polar, C_D vs. C_L . (Note $C_{L,max}$ 1.5 assumed)
- 5) What thrust force is generated by each engine in cruise (conditions of task 1) ($\sigma = 0$) at an aircraft mass of 59000 kg ?

Assumptions and Additional Data

- Location of max. relative thickness of airfoil: 39.90% of chord, from leading edge
- Sweep at max. relative thickness of airfoil: 25.78°

Surface	k [m]
Camouflaged aluminum	$10 \cdot 10^{-6}$
Flat coating	$6.3 \cdot 10^{-6}$
Unpolished sheet	$4.1 \cdot 10^{-6}$
Polished sheet	$1.5 \cdot 10^{-6}$
Flat CFRP	$0.5 \cdot 10^{-6}$

Equations

$$F_{fus} = \left(0.9 + \frac{5}{f^{1.5}} + \frac{f}{400}\right) \text{ with } f = \left(\frac{l}{d}\right)_{fus}$$

$$F_W = \left[1 + \frac{0.6}{(x/c)_m} \cdot \left(\frac{t}{c}\right)_W + 100 \cdot \left(\frac{t}{c}\right)_W^4\right] \cdot [1.34 M^{0.18} (\cos \Lambda_m)^{0.28}]$$

$$F_{nac} = \left[1 + 0.35/f\right] \text{ with } f = \left(\frac{l}{d}\right)_{nac}$$

$$S_{fus,wet} = \pi \cdot d^2 \cdot \left[\left(\frac{l}{d}\right)_{fus} - 1.35\right]$$

$$S_{W,wet} = 2 \cdot S_{ref}$$

$$S_{nac,wet} = \pi d_{nac} l_{nac}$$

$$Re_{CO,W} = 38 \cdot \left(\frac{l_{ref}}{k}\right)^{1.053} \text{ (subsonic); } Re_{CO} = 44 \cdot \left(\frac{l_{ref}}{k}\right)^{1.053} \cdot M_\infty^{1.16} \text{ (transsonic/supersonic)}$$

$$c_{f,lam} = \frac{1.328}{\sqrt{Re}}$$

$$c_{f,turb} = \frac{0.455}{(\log_{10} Re)^{2.58}} \cdot (1 + 0.144 \cdot M_\infty^2)^{-0.65}$$

$$\mu = 18.27 \cdot 10^{-6} \cdot \left(\frac{411.15}{T(^{\circ}K) + 120}\right) \cdot \left(\frac{T(^{\circ}K)}{291.15}\right)^{1.5}$$

$$f_{EMP} = C_{D,EMP} \cdot S_{EMP}$$

$$C_{L,PB} = 6.5 \cdot \frac{t}{c}$$

$$k = \frac{(C_L - C_{L,PB})}{(C_{L,max} - C_{L,PB})} \cdot \left(\frac{1}{C_{L\alpha}} - \frac{1}{\pi \cdot AR \cdot e}\right) + \frac{1}{\pi \cdot AR \cdot e}$$